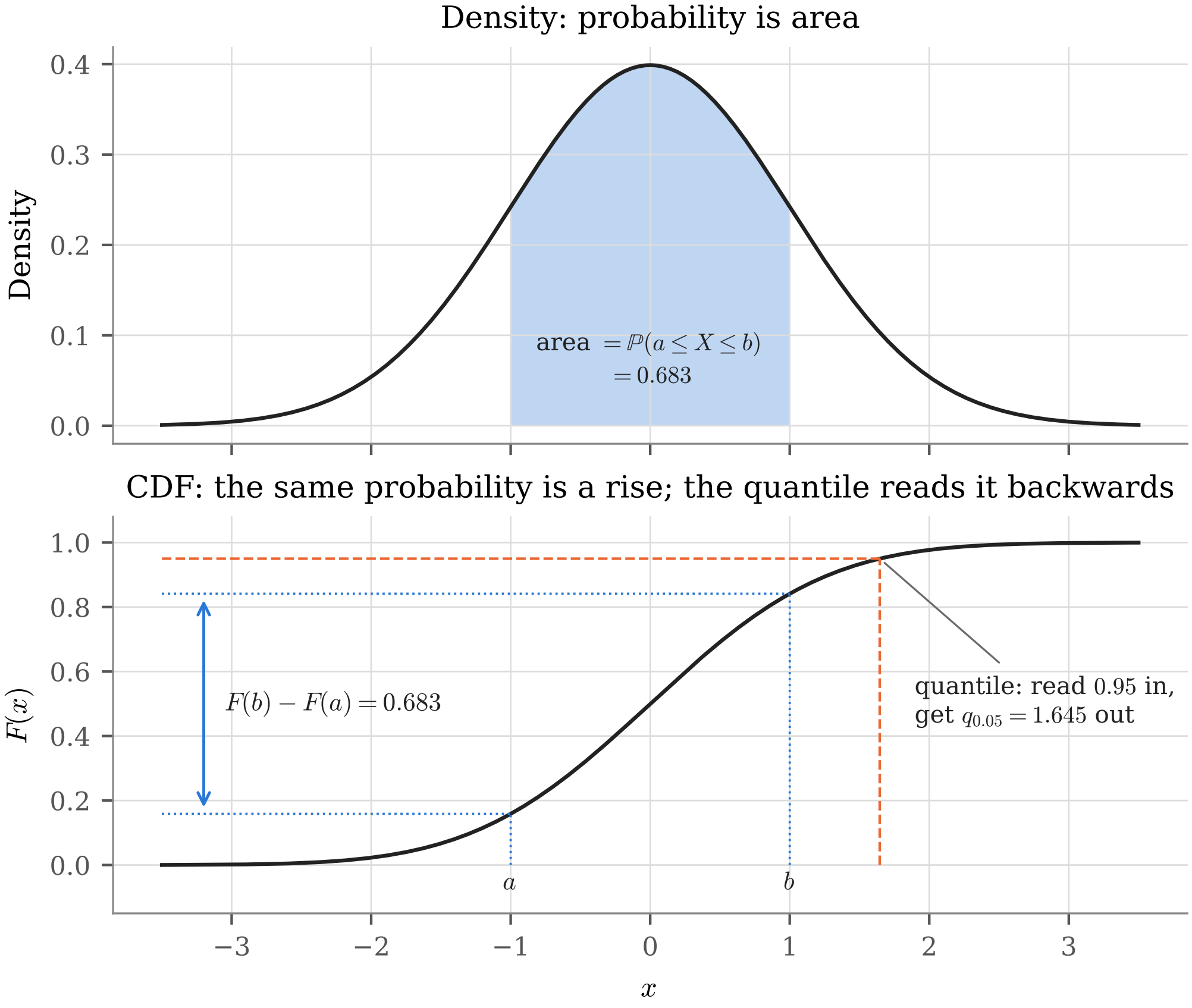
Density: probability is area
Density
0.4
0.3
0.2
0.1
0.0
area $= \mathbb{P}(a \le X \le b)$
$= 0.683$
CDF: the same probability is a rise; the quantile reads it backwards
$F(x)$
1.0
0.8
0.6
0.4
0.2
0.0
$F(b) - F(a) = 0.683$
quantile: read 0.95 in,
get $q_{0.05} = 1.645$ out
a
b
−3
−2
−1
0
1
2
3
x